8.8 Module Quiz

Date: 4/5/2026, 3:48:37 PM

Time Spent: 12:19

Score: 87%

Passing Score: 80%

Question 1 ✕ Incorrect

What is a golden configuration in network management?

- ○ The maximum configuration a device can handle
- ○ A template for the desired state of a device ✓ Correct
- ○ The configuration that achieves the highest speed
- ◉ The initial configuration of a device when it is first installed ✕ Incorrect

Explanation

A golden configuration, also known as a baseline configuration, serves as a template or standard for the optimal and secure configuration of a network device. It represents the approved settings and configurations that a device should have to operate effectively and securely within the network. By comparing current device configurations to the golden configuration, network administrators can quickly identify unauthorized changes, deviations, or misconfigurations, facilitating efficient management and compliance with security policies.

The golden configuration is not about the maximum configuration capacity but rather the ideal state for security and performance.

It is not specifically designed to achieve the highest speed but to ensure devices are configured optimally and securely.

The golden configuration is not necessarily the initial configuration but a standard that reflects the best practices and security policies of the organization.

Related Content

- 8.1.1 Configuration Management
- 8.1.2 Network Device Backup Management
- 8.2.5 Performance Monitoring

resources\questions\q_perf_mon_golden_configuration_n09.question.xml

Question 2 Correct

A security analyst is investigating a series of unusual network activities that suggest a potential breach in their organization's network.

The analyst suspects that the traffic is being routed through an unknown intermediary device within the network.

To map the path that the network traffic is taking to reach its destination, which Nmap command option should the security analyst use?

- nmap -sn 10.10.1.0/24
- nmap -A 10.10.1.0/24
- nmap --traceroute 10.10.1.0/24 ✓ Correct
- nmap -p 80,443 10.10.1.0/24

Explanation

The correct answer is the **nmap --traceroute 10.10.1.0/24** command. The --traceroute option in Nmap is specifically designed to trace the path packets take from the scanner to the target host. This would allow the security analyst to identify any unexpected devices or routes that the network traffic is passing through, which is crucial for investigating the potential breach.

The **-sn** option is used for host discovery without port scanning. While it can identify active hosts, it does not provide the path that packets take to reach these hosts, which is what the security analyst needs to investigate.

The **-A** option enables OS detection, version detection, script scanning, and traceroute. Although it includes traceroute, it also performs additional scans that may not be necessary for the security analyst's immediate need to map packet paths.

The **-p 80,443** option specifies that only ports 80 and 443 should be scanned. This command is useful for identifying web services but does not provide information on the path packets take, which is essential for the security analyst's investigation.

Related Content

- 7.2.9 Lab: Scan for Web Services with Nmap
- 8.2.2 Nmap
- 8.2.3 Nmap Port Scanning

resources\questions\q_nmap_traceroute_scenario_n09.question.xml

Question 3 Correct

A company's network administrator notices that during peak hours, the network's performance significantly drops, leading to slow file transfers and delayed email delivery.

What is the MOST likely cause of this issue?

- (●) Bottleneck ✓ Correct
- ( ) Excessive bandwidth
- ( ) Low jitter
- ( ) Efficient packet routing

Explanation

The correct answer is bottleneck. A bottleneck occurs when a specific part of the network becomes overloaded, reducing the overall network performance. During peak hours, the increased demand for network resources can lead to overutilization of certain devices or links, causing a bottleneck. This results in slower data transfer rates and delayed communications, as the network struggles to process the high volume of traffic.

Excessive bandwidth would actually mitigate performance issues by providing ample capacity for data transfer. It is unlikely to be the cause of reduced performance during peak hours.

Low jitter indicates a stable and consistent packet delivery time, which would contribute to better network performance, not issues.

Efficient packet routing would improve network performance by ensuring data takes the most optimal path. It is not a cause of performance degradation.

Related Content

- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_qos_metrics_bottleneck_scenario_n09.question.xml

Question 4 Correct

You are troubleshooting a network issue where certain packets are not reaching their intended destination. You want to capture all ICMP packets to analyze their paths and potential points of failure.

Which tcpdump command should you use to capture only ICMP packets?

- (selected) tcpdump -i any icmp ✓ Correct
- tcpdump -i eth0 -w icmp_packets.pcap
- tcpdump -i eth0 "protocol icmp"
- tcpdump -i eth0 "port 53"

Explanation

The correct command is **tcpdump -i any icmp**. This command captures all ICMP packets across all network interfaces (-i any), which is ideal for analyzing the paths of these packets. The icmp filter ensures that only ICMP packets are captured.

The **tcpdump -i eth0 -w icmp_packets.pcap** command writes packets to a file but lacks a filter for ICMP packets, capturing all traffic instead.

With the **tcpdump -i eth0 "protocol icmp"** command, the correct syntax for filtering ICMP packets is simply icmp, not protocol icmp.

The **tcpdump -i eth0 "port 53"** command captures traffic on port 53 (DNS) and does not target ICMP packets, which do not use a port number.

Related Content

- 1.3.7 Binary and Hexadecimal
- 2.3.5 Termination Tools and Techniques
- 4.1.3 Address Resolution Protocol
- 4.4.1 ipconfig
- 4.4.2 ifconfig and ip
- 4.4.3 arp
- 4.4.4 ping
- 4.4.5 Lab: IPv4 Troubleshooting Tools
- 4.4.6 Lab: IPv4 Troubleshooting tools for Linux

- 4.4.7 Lab: Use IPv4 Test Tools
- 4.6.1 IP Configuration Issues
- 4.6.2 Duplicate IP and MAC Address Issues
- 5.1.4 Packet Forwarding
- 5.1.7 Routing Table Tools
- 5.1.8 tracert and traceroute
- 5.1.10 Lab: Cisco Troubleshooting Tools
- 5.2.1 Dynamic Routing Protocols
- 5.3.2 Network Address Translation Types
- 5.3.3 Port Address Translation
- 5.6.1 Virtual LANs and Subnets
- 5.7.1 Routing Table Issues
- 5.7.2 Default Route and Routing Loop Issues
- 5.7.3 VLAN Assignment Issues
- 6.1.5 netstat
- 6.1.8 Lab: View Open Ports with netstat
- 6.2.1 DHCP Process
- 6.2.3 DHCP Options
- 6.4.6 Lab: Troubleshoot Address Pool Exhaustion
- 6.4.8 Lab: Explore DHCP Troubleshooting
- 6.4.9 Lab: Troubleshoot IP Configuration 1
- 6.4.10 Lab: Troubleshoot IP Configuration 2
- 6.4.11 Lab: Troubleshoot IP Configuration 3
- 6.5.5 Host Address and Canonical Name Records
- 6.5.15 Lab: Troubleshoot DNS Records
- 6.5.16 Configuring DNS Caching on Linux
- 6.6.1 Client DNS Issues

- 6.6.2 Name Resolution Issues
- 6.6.3 nslookup
- 6.6.4 dig
- 6.6.5 Lab: Explore nslookup
- 6.6.6 Lab: Use nslookup
- 7.1.1 Transport Layer Security
- 8.2.2 Nmap
- 8.5.2 tcpdump
- 8.6.1 Common Performance Issues
- 8.6.6 Traffic Shaping
- 9.2.2 Attack Types
- 9.3.1 On-Path Attacks
- 9.3.6 VLAN Hopping Attacks
- 9.4.4 DNS Attacks
- 10.1.5 Digital Certificates and PKI
- 10.3.5 Lab: Scan for Unsecure Protocols
- 10.4.5 Extensible Authentication Protocol and IEEE 802.1X
- 10.4.6 Port Guards
- 12.1.2 IEEE 802.11a and 5GHz Channel Bandwidth
- 12.1.3 IEEE 802.11b/g and 2.4GHz Channel Bandwidth
- 12.2.1 Infrastructure Network Type
- 12.4.3 Channel Overlap Issues
- 13.2.4 Internet Key Exchange
- 13.3.9 Lab: Use PowerShell Remote

resources\questions\q_tcdump_i_scenario_n09.question.xml

Question 5 Correct

A network administrator is monitoring a large network with multiple SNMP agents. They notice that the SNMP monitor is receiving a high volume of trap messages, indicating various notable events from different devices.

What action should the administrator take to ensure that the SNMP monitor can effectively manage and respond to these trap messages?

- ○ Disable all trap messages to reduce the load on the SNMP monitor.
- ◉ Configure the SNMP agents to send trap messages only for critical events. ✓ Correct
- ○ Increase the polling interval for all SNMP agents.
- ○ Switch to using community strings for device authentication.

Explanation

The correct answer is to configure the SNMP agents to send trap messages only for critical events. By configuring SNMP agents to send trap messages only for critical events, the administrator can reduce the volume of trap messages received by the monitor, allowing it to focus on significant alerts that require attention, thus improving the effectiveness of network monitoring and management.

Disabling all trap messages would prevent the SNMP monitor from receiving alerts about significant events, potentially leading to overlooked issues.

Increasing the polling interval would reduce the frequency of regular information updates from the SNMP agents but would not directly affect the volume of trap messages, which are event-driven.

Switching to using community strings for device authentication is a security configuration aspect and does not address the issue of managing a high volume of trap messages.

Related Content

- 8.2.1 Network Discovery
- 8.3.1 SNMP Agents and Monitors
- 8.3.2 SNMP Security
- 8.3.3 Configuring an SNMP System on a Router
- 8.3.4 Monitoring a Switch with SNMP
- 8.3.5 Configuring SNMP Trap

resources\questions\q_snmp_trap_messages_scenario_n09.question.xml

Question 6 Correct

What is the primary function of Cisco Discovery Protocol (CDP)?

- To encrypt data traffic between Cisco devices
- To assign IP addresses to devices on a network
- To discover information about directly connected Cisco devices ✓ Correct
- To provide a secure tunnel for data transmission

Explanation

The primary function of CDP is to discover information about directly connected Cisco devices. It allows devices to learn about each other, facilitating easier network management and troubleshooting.

Encrypting data traffic is not the function of CDP; it is typically handled by security protocols such as IPsec.

Assigning IP addresses to devices on a network is generally the role of DHCP, not CDP.

Providing a secure tunnel for data transmission is the function of VPN protocols, not CDP.

Related Content

5.6.5 Voice VLANs

8.2.4 Discovery Protocols

resources\questions\q_discovery_protocols_cdp_primary_role_n09.question.xml

Question 7 ✕ Incorrect

Why is it critical that users do not share logon details according to the document?

- To prevent the overloading of system resources
- To ensure accurate tracking of actions in the logs ✓ Correct
- To keep the system logs from being deleted
- To avoid duplicate entries in the application logs ✕ Incorrect

Explanation

The correct answer is to ensure accurate tracking of actions in the logs. Sharing login details undermines the integrity of the logs by making it impossible to accurately track actions back to the individual responsible, especially in the context of audit logs.

Overloading system resources is a concern but not related to the sharing of logon details.

The risk of logs being deleted is unrelated to the sharing of logon details.

Duplicate entries in application logs are not the primary concern with sharing logon details.

Related Content

- 8.4.1 Network Device Logs
- 8.4.2 Log Collectors and Syslog
- 8.4.3 Event Prioritization and Alerting
- 8.4.4 Security Information and Event Management
- 8.4.5 Log Reviews
- 8.4.6 Lab: Configure Logging in pfSense
- 8.4.7 Lab: Evaluate Event Logs in pfSense
- 8.4.8 Lab: Auditing Device Logs on a Cisco Switch
- 8.4.9 Lab: Configure Logging on Linux
- 8.4.10 Lab: View Event Logs

resources\questions\q_net_logs_logon_details_n09.question.xml

Question 8 Correct

A network administrator is tasked with improving the VoIP call quality within the company. They have identified packet loss as a significant issue.

Which of the following actions is MOST likely to reduce packet loss?

- Increasing the network's bandwidth
- Implementing QoS protocols ✓ Correct
- Upgrading the company's email server
- Decreasing the sampling frequency for audio

Explanation

The correct answer is implementing Quality of Service (QoS) protocols. Implementing QoS protocols is an effective way to manage packet prioritization, ensuring that real-time applications like VoIP receive the necessary bandwidth and minimal delay. QoS can help reduce packet loss by prioritizing voice traffic over less time-sensitive data, ensuring that VoIP calls maintain high quality even during periods of high network congestion.

While increasing bandwidth can alleviate some issues by providing more capacity, it does not address the prioritization of VoIP packets, which is crucial for reducing packet loss in real-time applications.

Upgrading the email server would not directly impact the packet loss experienced by VoIP calls, as it addresses a different aspect of the network's operations.

Decreasing the sampling frequency could reduce the quality of the audio in VoIP calls and does not address the issue of packet loss.

Related Content

- 7.3.4 VoIP Protocols
- 8.6.1 Common Performance Issues
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_qos_metrics_qos_protocols_scenario_n09.question.xml

Question 9 Correct

In Wireshark, what does the "Follow TCP Stream" context command do?

- Encrypts the TCP session for security purposes
- Reconstructs the packet contents for a TCP session ✓ Correct
- Disconnects the selected TCP session
- Increases the speed of the TCP session

Explanation

The "Follow TCP Stream" feature in Wireshark allows users to view the entire conversation between two endpoints in a TCP session by reconstructing the packet contents. This is useful for analyzing the data exchange in a session and troubleshooting issues.

The "Follow TCP Stream" feature does not encrypt the session; it is used for analysis purposes.

The "Follow TCP Stream" feature does not disconnect the session; it allows for the inspection of the session's data.

The "Follow TCP Stream" feature does not affect the speed of the session; it's a tool for analysis.

Related Content

- 4.6.2 Duplicate IP and MAC Address Issues
- 8.5.3 Protocol Analyzers
- 8.5.4 Using Wireshark to Troubleshoot Network Issues
- 8.5.5 Lab: Troubleshoot with Wireshark
- 8.6.1 Common Performance Issues
- 8.6.4 Traffic Testing Tools
- 9.3.7 Lab: Poison ARP and Analyze with Wireshark
- 9.5.2 Password Attacks

resources\questions\q_prot_analyze_follow_tcp_stream_n09.question.xml

Question 10 Correct

Which open-source tool is mentioned as a graphical packet capture and analysis utility?

- Nmap
- Wireshark ✓ Correct
- Nessus
- Netcat

Explanation

Wireshark is an open-source packet capture and analysis tool that provides a graphical interface for viewing and analyzing network traffic. It is widely used for network troubleshooting, analysis, software and protocol development, and education.

Nmap is primarily used for network discovery and security auditing.

Nessus is a vulnerability scanner used to identify vulnerabilities in networked systems.

Netcat is a networking utility for reading from and writing to network connections using TCP or UDP.

Related Content

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- 8.5.5 Lab: Troubleshoot with Wireshark
- 8.6.1 Common Performance Issues
- 8.6.4 Traffic Testing Tools
- 9.3.7 Lab: Poison ARP and Analyze with Wireshark
- 9.5.2 Password Attacks

resources\questions\q_prot_analyze_wireshark_n09.question.xml

Question 11

Correct

Which document captures the need or reasons for change and the procedure for implementing the change?

- ○ Standard Operating Procedure (SOP)
- ○ Service Request Ticket
- ◉ Request for Change (RFC) ✓ Correct
- ○ Change Advisory Board (CAB) document

Explanation

The Request for Change (RFC) document captures the need or reasons for change and the procedure for implementing the change, and it is submitted for approval. This document is central to the change management process.

SOP sets out the principal goals and considerations for performing a task, not specifically for implementing a change.

A service request ticket authorizes a configuration change but does not capture the need or reasons for change in detail.

The Change Advisory Board (CAB) may approve major changes but does not itself capture the need or reasons for change.

Related Content

resources\questions\q_agreements_rfc_n09.question.xml

Question 12 Correct

What is a benefit of supporting a version history of previous configurations on network devices?

- It allows for easy device replacement.
- It enables changes to be rolled back. ✓ Correct
- It reduces the need for backups.
- It simplifies network redesign.

Explanation

The correct answer is that it enables changes to be rolled back if they cause problems. Supporting a version history of previous configurations allows for changes to be easily rolled back if they cause operational issues. This feature enhances network stability and recovery capabilities.

While device replacement is an aspect of network management, version history specifically benefits configuration management and recovery.

Version history complements backups but does not reduce the need for them; both are important for comprehensive disaster recovery planning.

Simplifying network redesign is not directly related to the benefit of version history, which is more about managing and recovering from configuration changes.

Related Content

- 8.1.1 Configuration Management
- 8.1.2 Network Device Backup Management
- 8.1.6 Lifecycle Management

resources\questions\q_network_backup_version_history_n09.question.xml

Question 13 Correct

A network administrator is installing hundreds of Internet of Things (IoT) sensor devices at a corporate facility.

While each sensor device can be accessed and configured separately, the administrator requires a centralized remote management framework that can perform regular device polling as well as configurable thresholds for triggering notifications when events like port failure occur.

Which of the following is an agent-based framework the administrator can BEST fulfill these monitoring and remote management requirements?

- Secure Shell
- Remote Desktop Protocol
- Simple Network Management Protocol ✓ Correct
- Telnet

Explanation

The Simple Network Management Protocol (SNMP) is a widely used agent-based framework for remote management and monitoring of servers and network appliances.

Display-less devices such as routers are commonly configured remotely over the network by using a protocol such as Secure Shell (SSH).

Remote Desktop Protocol (RDP) is Microsoft's protocol for operating remote GUI connections to a Windows machine. RDP uses TCP port 3389.

Telnet is a tool that transmits shell commands and output between a client and a remote host. It was commonly used for remote management, but is considered deprecated and obsolete due to security concerns.

Related Content

- 8.2.1 Network Discovery
- 8.3.1 SNMP Agents and Monitors
- 8.3.2 SNMP Security
- 8.3.3 Configuring an SNMP System on a Router
- 8.3.4 Monitoring a Switch with SNMP
- 8.3.5 Configuring SNMP Trap

resources\questions\q_snmp_scenario_n09.question.xml

Question 14 Correct

What is the recommended practice when using SNMP v2c regarding community strings?

- Transmit them in plaintext over the network.
- Leave the community string set to the default.
- Use difficult to guess community strings. ✓ Correct
- Use SNMP v2c for secure environments only.

Explanation

When using SNMP v2c, it is recommended to use difficult to guess community strings and not to transmit them in plaintext over networks where they could be intercepted.

Transmitting community strings in plaintext over the network is a security risk because they can be intercepted, compromising SNMP security.

Leaving the community string set to the default or blank is a significant security risk, as it makes unauthorized access much easier for attackers.

While SNMP v2c lacks robust security features, it is still used in various environments; the focus should be on securing its use through practices like using complex community strings and ACLs.

Related Content

- 8.2.1 Network Discovery
- 8.3.1 SNMP Agents and Monitors
- 8.3.2 SNMP Security
- 8.3.3 Configuring an SNMP System on a Router
- 8.3.4 Monitoring a Switch with SNMP
- 8.3.5 Configuring SNMP Trap

resources\questions\q_snmp_v2_difficult_to_guess_n09.question.xml

Question 15 Correct

What is the purpose of using display filters in Wireshark?

- To increase the capture speed of packets
- To change the color of the interface
- To show only particular frames or sequences of frames ✓ Correct
- To delete unwanted packets from the network

Explanation

Display filters in Wireshark are used to narrow down the view to only specific frames or sequences of frames based on criteria set by the user. This is particularly useful when analyzing large amounts of data, as it allows the user to focus on the most relevant information.

Display filters do not affect the capture speed; they filter packets after they have been captured.

Changing the interface color is not the purpose of display filters; they are used for filtering data.

Display filters do not delete packets from the network; they only filter what is displayed in Wireshark.

Related Content

- 4.6.2 Duplicate IP and MAC Address Issues
- 8.5.3 Protocol Analyzers
- 8.5.4 Using Wireshark to Troubleshoot Network Issues
- 8.5.5 Lab: Troubleshoot with Wireshark
- 8.6.1 Common Performance Issues
- 8.6.4 Traffic Testing Tools
- 9.3.7 Lab: Poison ARP and Analyze with Wireshark
- 9.5.2 Password Attacks

resources\questions\q_prot_analyze_display_filters_n09.question.xml

Question 16 Correct

What role does the Spanning Tree Protocol (STP) play in preventing network issues?

- It increases network speed.
- It encrypts data transfers.
- It prevents switching loops. ✓ Correct
- It manages bandwidth allocation.

Explanation

The Spanning Tree Protocol (STP) is designed to prevent switching loops within a network. Switching loops occur when there are multiple paths between switches that can cause broadcast frames to circulate indefinitely, leading to broadcast storms and network failures. STP creates a loop-free logical network topology by blocking redundant paths, ensuring that data frames do not loop endlessly. This prevention of switching loops is crucial for maintaining network stability and performance.

STP's primary function is not to increase network speed but to prevent switching loops that can degrade network performance.

While securing data transfers is important, STP's role is focused on preventing switching loops, not on encrypting data or enhancing data transfer security.

Managing bandwidth allocation involves controlling how network resources are distributed among users and services. STP does not manage bandwidth allocation but prevents switching loops to ensure network stability.

Related Content

- 3.3.3 Spanning Tree Protocol
- 3.3.4 Spanning Tree Protocol Configuration
- 3.4.2 Port Status Indicators
- 3.4.3 Switch Show Commands
- 3.4.5 MAC Address Table
- 3.4.6 Network Loop and Broadcast Storm Issues
- 3.4.9 Lab: Switching Loop
- 8.2.5 Performance Monitoring
- 8.2.6 Availability Monitoring

9.3.4 MAC Flooding Attack

9.3.6 VLAN Hopping Attacks

resources\questions\q_perf_mon_stp_role_n09.question.xml

Question 17 Correct

An IT security manager has been alerted to a potential security breach in the company's network. An unknown user has been attempting to access restricted areas of the company's file server.

To track down the source of these unauthorized access attempts, the IT security manager decides to examine the relevant logs.

Which type of log should the IT security manager primarily focus on to identify the user behind these unauthorized access attempts?

- System log
- Application log
- Audit log ✓ Correct
- Performance/Traffic log

Explanation

The correct answer is audit logs. Audit logs are specifically designed to record success and failure events related to authentication and authorization. This makes them the ideal source for tracking unauthorized access attempts, as they can provide detailed information about who tried to access the system, when the attempt was made, and whether it was successful or not. By examining the audit logs, the IT manager can identify the user account used for the unauthorized attempts, aiding in the investigation and response to the security breach.

While system logs do record events at the OS level, including some security events, they are not as focused on authentication and authorization attempts as audit logs. System logs might not provide the detailed account-specific information needed.

Application logs are specific to individual applications and services, recording operational data. Unless the unauthorized access was to a specific application and that application maintains detailed access logs, application logs might not provide the necessary information about the unauthorized access attempts.

Performance and traffic logs track metrics for compute, storage, and network resources over time. While an unusual spike in traffic could indicate a security issue, these logs would not directly identify unauthorized access attempts or the user behind them.

Related Content

- 8.4.1 Network Device Logs
- 8.4.2 Log Collectors and Syslog
- 8.4.3 Event Prioritization and Alerting

8.4.4 Security Information and Event Management

8.4.5 Log Reviews

8.4.6 Lab: Configure Logging in pfSense

8.4.7 Lab: Evaluate Event Logs in pfSense

8.4.8 Lab: Auditing Device Logs on a Cisco Switch

8.4.9 Lab: Configure Logging on Linux

8.4.10 Lab: View Event Logs

resources\questions\q_net_logs_audit_log_scenario_n09.question.xml

Question 18 Correct

What are latency and jitter in the context of network performance?

- Types of network security protocols
- Measures of network speed and efficiency
- Problems of timing and sequence of packet delivery ✓ Correct
- Techniques for optimizing bandwidth usage

Explanation

Latency refers to the delay before a transfer of data begins following an instruction for its transfer, while jitter is the variation in the delay of received packets. Both are crucial in determining the quality of real-time applications like VoIP and video streaming, as they affect the smoothness and quality of the transmission.

Latency and jitter are not security protocols but are related to the performance of data transmission.

While they can affect network speed and efficiency, latency and jitter specifically refer to timing and sequence issues in packet delivery.

These are not techniques for optimizing bandwidth usage but are issues that can negatively impact the quality of network services.

Related Content

8.6.1 Common Performance Issues

resources\questions\q_qos_metrics_latency_and_jitter_description_n09.question.xml

Question 19 Correct

How can state information be preserved if it needs to be backed up?

- ○ By performing a factory reset
- ◉ By configuring the appliance to log state data ✓ Correct
- ○ By manually documenting the state information
- ○ By using the default configuration

Explanation

If state information needs to be preserved, the appliance should be configured to log state data to a remote server, using a protocol such as syslog. This ensures that crucial operational data is backed up and can be restored if necessary.

Performing a factory reset would erase all configurations and state information, not preserve it.

Manually documenting state information is impractical and error-prone for dynamic and detailed operational data.

Using the default configuration does not address the need to back up current operational state information.

Related Content

- 8.1.1 Configuration Management
- 8.1.2 Network Device Backup Management
- 8.1.6 Lifecycle Management

resources\questions\q_network_backup_log_state_data_n09.question.xml

Question 20 Correct

What is an OID in the context of SNMP?

- (●) Object Identifier ✓ Correct
- ( ) Operational Interface Descriptor
- ( ) Optical Interference Device
- ( ) Original Internet Domain

Explanation

In SNMP, an OID (Object Identifier) refers to a numeric identifier used to uniquely identify each parameter stored in a MIB.

Operational Interface Descriptor sounds technical but is not related to SNMP, which uses OIDs (Object Identifiers) to uniquely identify MIB parameters.

Optical Interference Device pertains to optical technology and has no connection to SNMP, which uses OIDs for parameter identification in network management.

Original Internet Domain is unrelated to SNMP's use of OIDs, which are numeric identifiers for parameters in a device's MIB, not domain names.

Related Content

- 8.2.1 Network Discovery
- 8.3.1 SNMP Agents and Monitors
- 8.3.2 SNMP Security
- 8.3.3 Configuring an SNMP System on a Router
- 8.3.4 Monitoring a Switch with SNMP
- 8.3.5 Configuring SNMP Trap

resources\questions\q_snmp_oid_role_n09.question.xml

Question 21 Correct

You are a network administrator for a large organization. One morning, you receive several alerts from the automated event management system. Among these alerts are notifications of packet loss exceeding 5%, a critical system service failure, and multiple failed login attempts on a key server.

Given the need to prioritize your response, which alert should you address first?

- Packet loss exceeding 5%
- Critical system service failure ✓ Correct
- Multiple failed login attempts on a key server
- All alerts should be addressed simultaneously

Explanation

A critical system service failure can have immediate and widespread impact on the organization's operations, potentially affecting all users and critical processes. Addressing this first ensures that essential services are restored to maintain business continuity.

While packet loss exceeding 5% can indicate significant network issues, it may not immediately impact critical services or security as directly as a critical system service failure.

Multiple failed login attempts could indicate a security threat, which is important but may not have the immediate operational impact of a critical system service failure. It should be prioritized accordingly after addressing any direct operational threats.

While all alerts are important, resources are often limited, and alerts must be prioritized based on their impact. Addressing all alerts simultaneously is not feasible in most scenarios.

Related Content

- 8.4.1 Network Device Logs
- 8.4.2 Log Collectors and Syslog
- 8.4.3 Event Prioritization and Alerting
- 8.4.4 Security Information and Event Management
- 8.4.5 Log Reviews
- 8.4.6 Lab: Configure Logging in pfSense
- 8.4.7 Lab: Evaluate Event Logs in pfSense
- 8.4.8 Lab: Auditing Device Logs on a Cisco Switch

8.4.9 Lab: Configure Logging on Linux

8.4.10 Lab: View Event Logs

resources\questions\q_event_alert_critical_failure_scenario_n09.question.xml

Question 22 Correct

Why is plotting data as a graph helpful in log analysis?

- It eliminates the need for software analysis.
- It makes it easier to spot trends or spikes in the data. ✓ Correct
- It reduces the amount of data that needs to be stored.
- It increases the speed of the network.

Explanation

Visual representations of data, such as graphs, make it significantly easier to identify trends, spikes, or troughs. This visual approach allows analysts to quickly spot patterns or anomalies that might be difficult to discern in raw data, facilitating faster and more effective decision-making.

Plotting data as a graph does not eliminate the need for software analysis; it is a tool used within the analysis process to aid in understanding the data.

The method of data visualization does not impact the amount of data that needs to be stored. Data storage requirements are determined by other factors.

The speed of the network is not directly affected by how data is plotted or analyzed in log reviews. Visualization aids in analysis, not network performance.

Related Content

- 8.4.1 Network Device Logs
- 8.4.2 Log Collectors and Syslog
- 8.4.3 Event Prioritization and Alerting
- 8.4.4 Security Information and Event Management
- 8.4.5 Log Reviews
- 8.4.6 Lab: Configure Logging in pfSense
- 8.4.7 Lab: Evaluate Event Logs in pfSense
- 8.4.8 Lab: Auditing Device Logs on a Cisco Switch
- 8.4.9 Lab: Configure Logging on Linux
- 8.4.10 Lab: View Event Logs

resources\questions\q_log_review_plotting_data_n09.question.xml

Question 23 Correct

What role do connectors or plug-ins play in SIEM software?

- They increase the physical security of the SIEM server.
- They interpret data from distinct types of systems. ✓ Correct
- They serve as the primary user interface for SIEM software.
- They encrypt log data for secure transmission.

Explanation

The correct answer is that they interpret data from distinct types of systems for log aggregation. Connectors or plug-ins in SIEM software are designed to interpret (or parse) data from distinct types of systems and account for differences between vendor implementations. This is crucial for the log aggregation process, as it enables the SIEM tool to understand and normalize data from various sources, making it possible to analyze the data collectively.

Connectors or plug-ins are not related to the physical security of the SIEM server; they are software components that help in data interpretation.

While connectors or plug-ins are essential for data processing, they do not serve as the primary user interface for SIEM software. The user interface is a separate component designed for interaction with the user.

The role of connectors or plug-ins is to interpret and normalize data, not to encrypt it. Encryption of log data may occur at different stages for security purposes, but it is not the primary function of connectors or plug-ins.

Related Content

- 8.4.1 Network Device Logs
- 8.4.2 Log Collectors and Syslog
- 8.4.3 Event Prioritization and Alerting
- 8.4.4 Security Information and Event Management
- 8.4.5 Log Reviews
- 8.4.6 Lab: Configure Logging in pfSense
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8.4.10 Lab: View Event Logs

resources\questions\q_siem_connectors_role_n09.question.xml

Question 24

Correct

Why might frames with errors not be mirrored in a SPAN/port mirroring setup?

- Because the switch is designed to correct errors automatically
- Because frames with errors are encrypted
- Because frames with errors will not be mirrored and may be dropped ✓ Correct
- Because the sniffer software filters out errors

Explanation

In SPAN/port mirroring, the switch may not mirror frames with errors or may drop frames under heavy load due to limitations in its mirroring capabilities. This can result in incomplete data capture for analysis.

While switches can detect and sometimes correct errors, this does not explain why frames with errors might not be mirrored.

Encryption does not affect whether frames with errors are mirrored. Encryption is about data privacy, not error handling.

The decision to mirror frames, including those with errors, is made by the switch hardware before the sniffer software has a chance to apply any filters.

Related Content

8.5.1 Packet Capture

8.5.6 Lab: Configure Port Mirroring

10.4.8 Port Mirroring

resources\questions\q_sniffers_no_mirroring_n09.question.xml

Question 25 ✕ Incorrect

Which Nmap scan type is less stealthy due to its use of the operating system to attempt a full TCP connection?

- (●) TCP SYN (-sS) ✕ Incorrect
- ( ) TCP connect (-sT) ✓ Correct
- ( ) UDP scans (-sU)
- ( ) Port range (-p)

Explanation

The TCP connect scan (-sT) uses the operating system to attempt a full TCP connection, making it less stealthy because it completes the TCP handshake, which is more likely to be logged by the target system.

TCP SYN (-sS) is more stealthy because it does not complete the TCP handshake.

UDP scans (-sU) involve UDP ports and do not attempt a full TCP connection.

Specifying a port range (-p) is an option that can be used with various scan types and does not pertain to the method of scanning.

Related Content

- 7.2.9 Lab: Scan for Web Services with Nmap
- 8.2.2 Nmap
- 8.2.3 Nmap Port Scanning

resources\questions\q_port_scanner_st_purpose_n09.question.xml

Question 26 Correct

What is the primary purpose of Quality of Service (QoS) protocols and appliances?

- To increase the bandwidth of the network
- To support real-time services ✓ Correct
- To reduce the cost of network infrastructure
- To eliminate the need for packet switching

Explanation

QoS protocols and appliances are designed specifically to support real-time services such as voice and video communications, which have stringent requirements for timely and orderly delivery of data. This is crucial for maintaining the quality of these services.

Increasing the bandwidth is a way to improve network performance but is not the primary purpose of QoS. QoS focuses on managing and prioritizing network traffic.

Reducing the cost of network infrastructure is a goal for many organizations, but it is not the purpose of QoS protocols and appliances.

Packet switching is a fundamental method used in network communications, and QoS works within this framework to prioritize certain types of traffic, not to eliminate packet switching.

Related Content

- 7.3.4 VoIP Protocols
- 8.6.1 Common Performance Issues
- 8.6.5 Bandwidth Management
- 8.6.6 Traffic Shaping

resources\questions\q_qos_metrics_primary_purpose_n09.question.xml

Question 27 Correct

How does an SNMP monitor typically retrieve information from a device?

- By using the Set command
- By using the Get command ✓ Correct
- By establishing a VPN connection
- By initiating an SSH session

Explanation

The Get command is used by an SNMP monitor to query an agent for a single OID, which is a common method for retrieving information from devices.

The Set command is used by SNMP monitors to change the value of a variable in a device's MIB, not for the primary retrieval of information.

VPN connections are used to securely connect remote networks over the Internet, not for the SNMP monitor's task of querying devices for information.

SSH sessions provide secure shell access to devices for command-line interaction, differing from SNMP's method of polling devices using the Get command.

Related Content

- 8.2.1 Network Discovery
- 8.3.1 SNMP Agents and Monitors
- 8.3.2 SNMP Security
- 8.3.3 Configuring an SNMP System on a Router
- 8.3.4 Monitoring a Switch with SNMP
- 8.3.5 Configuring SNMP Trap

resources\questions\q_snmp_get_command_n09.question.xml

Question 28 Correct

What is the primary purpose of using traffic analysis tools in network performance testing?

- To increase the network's data rate
- To simulate traffic loads and observe performance ✓ Correct
- To physically inspect network cables
- To enhance the graphical interface of network applications

Explanation

Traffic analysis tools are used to simulate traffic loads on a network to observe and measure its performance under various conditions. This helps in identifying and addressing potential performance issues without the need to directly observe poor performance, making it an essential practice for maintaining network efficiency.

Increasing the network's data rate is a potential outcome of optimizing network performance, not the primary purpose of using traffic analysis tools.

Physical inspection of network cables is a method for identifying hardware issues, not for simulating traffic loads or measuring network performance.

Enhancing the graphical interface of network applications is unrelated to the purpose of traffic analysis tools, which focus on performance testing rather than user interface improvements.

Related Content

- 8.4.1 Network Device Logs
- 8.5.1 Packet Capture
- 8.6.1 Common Performance Issues
- 8.6.2 Interface Statistics
- 8.6.3 Flow Data
- 8.6.4 Traffic Testing Tools
- 8.6.5 Bandwidth Management

resources\questions\q_traffic_analyze_primary_purpose_n09.question.xml

Question 29 Correct

What is the primary reason for making a backup before performing a firmware update?

- To increase the device's storage capacity
- To prevent data loss in case the update fails ✓ Correct
- To speed up the update process
- To comply with legal requirements

Explanation

Making a backup ensures that the original system configuration can be restored if the update process encounters problems, preventing data loss and system malfunction.

Backups do not affect the device's storage capacity; they are a precautionary measure.

Backups do not influence the speed of the update process.

While backups may help with compliance, the primary reason is to safeguard against data loss during updates.

Related Content

8.1.6 Lifecycle Management

8.1.12 Lab: Update Firmware

resources\questions\q_life_cycle_firmware_update_n09.question.xml

Question 30 ✕ Incorrect

Where can standard network icons created by Cisco be obtained?

- ○ Cisco's official website ✓ Correct
- ○ Microsoft Office Online
- ◉ Adobe Stock ✕ Incorrect
- ○ AutoCAD Library

Explanation

Cisco's standard network icons, which are recognized as standards for network diagrams, can be obtained freely (without alteration) from Cisco's official website, providing a consistent and professional way to represent network devices.

Microsoft Office Online offers a variety of templates and icons but not the specific standard network icons created by Cisco.

Adobe Stock is a platform for stock images, illustrations, and graphics, not specifically for standard network icons by Cisco.

The AutoCAD Library contains CAD drawings and models for engineering and architecture, not standard network icons by Cisco.

Related Content

8.1.8 Physical Network Diagrams

8.1.9 Logical Network Diagrams

resources\questions\q_logi_diagram_network_icons_n09.question.xml

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